Assignment 18: Factoring Quadratics
DO NOT WRITE ON THIS PAPER

Part A

Use the following equation to fill out the t-table below.

$$y = x^2 + 8x + 12$$

x	y
-6	
-5	
-4	
-3	
-2	

Part B

Solve the quadratic equation $x^2 + 8x + 12 = 0$ and show your work.

Do you recognize any values from the T-table in part A? If so what values and what are special about them?

Part C

Factor the following quadratic expressions and show your work.

1. Factor by grouping: $5y^2 + 12y - 9$
2. Difference of Squares: $49w^2 - 4$

Question 1

Big Question: What does it mean to factor a quadratic completely? Why do we try and factor to solve quadratic equations?

Use the phrases “Zero-Product Property, Linear Terms, and Product” in your answer.

Then factor the following quadratic expression completely and show your work: $3x^2 - 2x - 21$

Question 2 Work	Question 2 Answer
If I factor a quadratic expression how can I check my work? If I solve a quadratic equation how can I check my work? *Hint: Think about the example from the slides:* $x^2 - 3x + 2$.	Write your answer in your Class Notes

Question 3 Work	Question 3 Answer
Solve $2x^2 + x - 21 = 0$. Show your work and check your answer.	Write your answer in your Class Notes

Additional Space